

Kasra Ferdowsi

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Research Interests

My research is at the intersection of *Human-Computer Interactions* and *Programming Languages*. Specifically, I hope to bridge the gap between cutting-edge academic research in programming languages and everyday programming by designing and building tools and interaction models that are useful to, and usable by, people who write code.

Education

UC San Diego	Sep 2019 - Current
- PhD in Computer Science and Engineering - Depth Area: Programming Languages, Compilers, and Software Engineering - Expected Graduation Date: June 2024	
UC Berkeley	Jun 2017 - May 2019
- B.S. in Electrical Engineering and Computer Science - Graduated with Highest Honors	
Diablo Valley College	Aug 2014 - May 2017
- A.S. for Transfer in Mathematics - A.S. for Transfer in Physics	

Publications

ColDeco: An End User Spreadsheet Inspection Tool for AI-Generated Code	VL/HCC 2023
Ferdowsi, K., Williams, J., Drosos, I., Gordon, A.D., Negreanu, C., Polikarpova, N., Sarkar, A., and Zorn, B.	
Investigating the Impact of Using a Live Programming Environment in a CS1 Course	SIGCSE TS 2022
Huang, R., Ferdowsi, K., Selvaraj, A., Soosai Raj, A.G., and Lerner, S.	
LooPy: Interactive Program Synthesis with Control Structures	OOPSLA 2021
Ferdowsifard, K., Barke, S., Peleg, H., Lerner, S., and Polikarpova, N.	
Small-Step Live Programming by Example	UIST 2020
Ferdowsifard, K., Ordokhanians, A., Peleg, H., Lerner, S., and Polikarpova, N.	

Research Experience

Research Internship Project at Microsoft Research	2023
- Ongoing research.	
LEAP: Live Exploration of AI-Generated Programs	2023
- With Ruanqianqian (Lisa) Huang, Michael B. James, Sorin Lerner, and Nadia Polikarpova. - Designed and implemented a novel interaction model for AI programming assistants that support exploring multiple code suggestions through Live Programming. - Ran a 17 participant between-subjects user study showing numerous benefits to this interaction model, including preventing over-reliance, and reducing cognitive load. - Link to ArXiv submission: https://arxiv.org/abs/2306.09541	
Towards Human-Centered Types and Type Debugging	2022
- PLATEAU'23 Workshop paper and presentation. - Surveyed the history of the usability of type systems, and type error messages. And offered concrete future directions based on cutting-edge HCI research. - Link to publication: https://doi.org/10.1184/R1/22227457.v1	

- ColDeco: An End-User Spreadsheet Inspection Tool for AI-Generated Code** 2022
- With Jack Williams, Ian Drosos, Andrew Gordon, Carina Negreanu, Advait Sarkar, and Ben Zorn.
 - Designed and implemented a novel interaction model for inspecting and diagnosing errors in AI-generated code, targeted at end-users.
 - Conducted a 24-participant controlled experiment to evaluate the interaction.
- The Usability of Advanced Type Systems** 2022
- PhD Research Exam.
 - Surveyed publications on advanced type systems which inspired the Rust programming language, as well as all usability studies on Rust itself.
 - Link to the written report: https://weirdmachine.me/papers/usability_of_advanced_type_systems.pdf
- Loopy: Interactive Program Synthesis with Control Structures** 2020 - 2021
- With Shraddha Barke, Hila Peleg, Sorin Lerner, and Nadia Polikarpova.
 - Designed and implemented a novel technique for using Bottom-up Enumerative Synthesis for efficient synthesis of blocks of Python code.
 - Links to the online editor, the paper, and conference talk: <http://loopy.goto.ucsd.edu/>
- Impact of Live Programming in a CS1 Course** 2020 - 2021
- With Ruanqianqian Huang, Ana Selvaraj, Adalbert Gerald Soosai Raj and Sorin Lerner.
 - Implemented a web editor for the Live Programming environment *Projection Boxes*, and developed and deployed a server for collecting usage data from the editor.
 - Ran a user study of hundreds of students in a CS1 course, and visualized and performed statistical analysis on data collected from the study.
- SnipPy: Small-Step Live Programming by Example** 2020
- With Allen Ordoookhanians, Hila Peleg, Sorin Lerner, and Nadia Polikarpova.
 - Designed and implemented a novel Program Synthesis interaction model with *Projection Boxes*.
 - Ran a 12-participant user study for evaluating the interaction.
 - Links to the online editor, the paper, and conference talk: <https://snippy.goto.ucsd.edu/>

Teaching Experience

- Teaching Assistant CSE230: Programming Languages** Fall 2022
- Held weekly office hours, graded assignments, and assisted students on Piazza.
- Teaching Assistant CSE130: Programming Languages** Winter 2022
- Presented discussion sections, held weekly office hours, and assisted students with technical and conceptual problems on Piazza.
- Teaching Assistant CSE130: Programming Languages** Fall 2021
- Presented discussion sections, held weekly office hours, and assisted students with technical and conceptual concepts on Piazza.
- Teaching Assistant CSE11: Accelerated Intro to Programming** Spring 2020
- Took responsibility for one week of the course in the instructor's absence, leading 2 lectures (in a live Q&A format with pre-recorded videos), and giving an online midterm exam to over 200 students.
 - Led the weekly discussion sections, including writing review notes and practice problems tailored to the course, leading live discussions, and creating pre-recorded videos.
- Teaching Assistant CSE131: Compiler Construction** Fall 2019
- Created autograder tests and infrastructure for the class projects.
 - Updated existing projects from Intel x86 to x86_64
 - Co-led the midterm review discussion section.

Service

- Program Committee Member at HATRA'23** 2023
- Student Volunteer at PLDI'23** 2023

Mentorship

GradWIC Mentor	2022 - 2023
CSE599: Teaching Methods in Computer Science Mentor	Winter 2023
ERSP Mentor	2021 - 2022
- With Ruanqianqian (Lisa) Huang, Nadia Polikarpova and Sorin Lerner - Link to poster: https://drive.google.com/file/d/1HZx_FODLaYjbIDEKttTFO5jsF8EUov_Y/	

Engineering Experience

Engineering Intern <i>Tom Sawyer Software</i>	Jun 2021 - Sep 2021
- Designed and developed the <i>Deploy to Kubernetes</i> feature of <i>Tom Sawyer Innovator</i>. - Lead the effort to document and improve <i>Innovator's</i> user and development experience.	
Engineering Intern <i>Tom Sawyer Software</i>	Apr 2020 - Jun 2020
- Co-designed and developed the <i>Tom Sawyer Application Innovator</i> (work in progress), a web application using and managing Git, Maven and Docker to streamline the build and deployment of Tom Sawyer Perspectives applications. - Main contributions included user-interface additions and improvements, an overhaul of the version control system, and bug fixes and improvements.	
Engineering Intern <i>Tom Sawyer Software</i>	May 2016 - May 2019
- Helped design and develop the <i>Tom Sawyer Licensing Server 2.0</i>, a backwards-compatible scalable web server using a microservice architecture. - Implemented secure and user-friendly account management features, and a responsive web interface for <i>Tom Sawyer Graph Database Browser</i>, as well as database organization and query improvements, and other fixes.	